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MapC ICETASK3 ST10434066

Q2.1.1) $|2x-1|-7 \geq -3$

$|2x-1| \geq 4$

$2x-1 \geq -4$ or $2x-1 \geq 4$

$2x \geq -3$ or $2x \geq 5$

$x \geq \frac{-3}{2}$ $x \geq \frac{5}{2}$

$\therefore -\frac{3}{2} \leq x \leq \frac{5}{2}$

Q2.1.2) $|x+4|-6 < 9$

$|x+4| < 15$

$x+4 < 15$ or $x+4 < -15$

$x < 11$ or $x < -19$

$\therefore -19 < x < 11$

Q2.2.1) $\frac{x+2}{x^2+2x-15}$

$Df = x^2+2x-15 = 0$

$x^2+5x-3x-15=0$

$x(x+5)-3(x+5)=0$

$x=0$ or $x=3$ or $x=-5$

$\therefore Df = \{x \mid x \in \mathbb{R}, x \neq 0, 3, -5\}$

Q2.2.2) $\frac{4x}{5x^2-30x+25}$

$5x^2-30x+25=0$

$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$\frac{-30 \pm \sqrt{30^2-4(5)(25)}}{2(5)}$

$= \frac{-30 \pm \sqrt{900-500}}{10}$

$= \frac{-30 \pm \sqrt{400}}{10}$

$= \frac{-30 \pm (20)}{10}$

$= \frac{-30+20}{10}$ or $\frac{-30-20}{10}$

$= \frac{-10}{10} = -1$ or $\frac{-50}{10} = -5$

$D_f = \{x \mid x \in \mathbb{R}, x \neq -1 \text{ or } -5\}$

Q2.3) $3|x-1| - 1 = 11$

$|x-1| - \frac{1}{3} = \frac{11}{3}$

$|x-1| = 4$

$x - 1 = -4$ or $x - 1 = 4$

$x = -3$ or $x = 5$

Q2.4.1

Q2.4.1) $\frac{x^2 - 4}{x^2 + 4x + 4}$

$= \frac{(x-2)(x+2)}{(x+2)^2}$

$$= \frac{3x-2}{x+2} \longrightarrow$$

$$\frac{9x^2 - 25y^2}{3x^2 - 5xy}$$

$$= \frac{(3x+5y)(3x-5y)}{x(3x-5y)}$$

$$= \frac{3x+5y}{x} \longrightarrow$$

Q2.4.3) $$\frac{64a^3 - 125b^3}{4a^2b - 5ab^2}$$

$$= \frac{(4a-5b)(16a^2+20ab+25b^2)}{ab(4a-5b)}$$

$$= \frac{16a^2+20ab+25b^2}{ab}$$

Q2.4.4) $\sqrt{4x^2y^4} \cdot \sqrt[3]{27x^3y^6}$

$2xy^2 \cdot 3xy^3$

$= 6x^2y^5$ →

Q2.4.5) $[xy^2]^4 \cdot \left[\sqrt[4]{x^3y^5}\right]^{-4}$

$= x^4y^8 \cdot [x^{\frac{3}{4}}y^{\frac{5}{4}}]^{-4}$

$= x^4y^8 \cdot x^{-3}y^{-5}$

$= xy^3$ ⇒